

# MODULAR FORMS AND $\ell$ -ADIC REPRESENTATIONS

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**(3.13)** Let  $p$  be a prime number, let  $n \geq 3$  be an integer prime to  $p$ , and let  $F_{n,p}$  be the functor which associates to each scheme  $S$  the set of isomorphism classes of commutative diagrams of  $S$ -schemes

$$\begin{array}{ccc} & & \mathbb{Z}/(n)^2 \\ & \nearrow \alpha & \nwarrow \alpha' \\ E_n & \xrightarrow{\quad} & F_n \\ \cap & & \cap \\ E & \xrightarrow{\varphi} & F \end{array} \quad (3.14)$$

where  $\varphi$  is a  $p$ -isogeny between elliptic curves and  $\alpha$  is an isomorphism. One denotes by  $q_1$  and  $q_2 : F_{n,p} \rightarrow M_n$  the morphisms of functors associating to a diagram (3.14) the subdiagrams  $(E, E_n, \alpha)$  and  $(F, F_n, \alpha')$ .

**Proposition (3.15).** *The functor  $F_{n,p}$  is represented by a scheme  $M_{n,p}$ , and the morphisms  $q_1, q_2 : M_{n,p} \rightarrow M_n$  are finite.*

The automorphism  $\sigma$  of  $F_{n,p}$  sending  $\varphi : E \rightarrow F$  to  ${}^t\varphi : F \rightarrow E$  exchanges  $q_1$  and  $q_2$ ; it suffices therefore to consider  $q_1$ . This morphism identifies  $F_{n,p}$  with the functor of subgroups of order  $p$  of the universal elliptic curve  $E$  over  $M_n$ , so that, by the theory of Hilbert schemes,  $F_{n,p}$  is representable and  $M_{n,p}$  is proper over  $M_n$ . If  $s$  is a geometric point of  $M_n$ ,  $q_1^{-1}(s)$  is the set of subgroups of order  $p$  of  $E_s$ , and has  $p+1$  elements if  $\text{char}(k(s)) \neq p$ , and only one, the kernel of Frobenius, if  $\text{char}(k(s)) = p$ .  $\square$

One can show that  $M_{n,p}$  is regular, and that  $q_1$  and  $q_2$  are finite flat; we do not use this delicate result, but content ourselves here to note that over  $\text{Spec}(\mathbb{Z}[1/p])$ , each  $q_i$  makes  $M_{n,p}$  an étale covering of degree  $p+1$  of  $M_n$ .

The morphisms  $q_i$  can be inserted into a commutative diagram

$$\begin{array}{ccccc} q_1^* E & \xrightarrow{\varphi} & q_2^* E & & \\ \swarrow & & \searrow & & \\ E & \xleftarrow{u} & M_{n,p} & \xrightarrow{v} & E \\ \downarrow f_n & & \downarrow q_1 \quad \downarrow q_2 & & \downarrow f_n \\ M_n & & M_n & & M_n \end{array} \quad (3.16)$$

where  $(\varphi, u, v)$  is part of the universal diagram (3.14).

One denotes by  $I_p$  the morphism from  $M_n$  to  $M_n$  corresponding to the morphism of functors  $(E, \alpha) \mapsto (E, \alpha/p)$ :

$$\begin{array}{ccc} E & \longrightarrow & E \\ \downarrow & & \downarrow \\ M_n & \xrightarrow{I_p} & M_n \end{array} \quad I_p^*(E, \alpha) = (E, \alpha/p). \quad (3.17)$$

$I_p^*$  is an automorphism of  $\tilde{H}^i(M_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \mathbb{Z}))$ . It is tiresome, but routine, to prove the following.

**Proposition (3.18).**

(i) *The endomorphism  $T_p$  of  ${}_n W$  is expressed, with the aid of (3.16), as the composite map*

$$\begin{aligned} \tilde{H}^1(M_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \mathbb{Q})) &\xrightarrow{q_2^*} \tilde{H}^1(M_{n,p}^{\text{an}}, \text{Sym}^k(R^1 v_* \mathbb{Q})) \\ &\xrightarrow{\varphi^*} \tilde{H}^1(M_{n,p}^{\text{an}}, \text{Sym}^k(R^1 u_* \mathbb{Q})) \xrightarrow{q_{1*}} \tilde{H}^1(M_n^{\text{an}}, \text{Sym}^k(R^1 f_{n*} \mathbb{Q})). \end{aligned}$$

where  $q_{1*}$  is the trace morphism for the covering  $q_1$ .

(ii) *Similarly,  $R_p = p^k I_p^*$ .*

$\square$

The distrustful reader could forget the adelic preliminaries and define  $T_p$  by (i). When  $n = 1$  or  $2$ , one puts  ${}_n W = W^{K_n}$ , so that

$${}_1 W = {}_n W^{\text{GL}_2(\mathbb{Z}/(n))}.$$

If  $S_{k+2}$  denotes the space of cusp forms, for the group  $\mathrm{SL}_2(\mathbb{Z})$ , of weight  $k+2$ , the Shimura isomorphism (3.11) induces an isomorphism

$${}^k_1W_\infty = {}^k_1W \otimes \mathbb{C} = S_{k+2} \oplus \bar{S}_{k+2}.$$

It is tiresome, but routine, to prove the following.

**Proposition (3.19).** *The endomorphism  $T_p$  of  ${}^k_1W_\infty$  is identified, via the Shimura isomorphism, with the direct sum of the Hecke operator on  $S_{k+2}$ , including the factor  $p^{k-1}$ , and its conjugate.*

□

(3.20) One has canonically

$$\bigwedge^2 R^1 f_{n*} \mathbb{Z}_\ell \simeq R^2 f_{n*} \mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(-1),$$

so that  $\mathrm{Sym}^k(R^1 f_{n*} \mathbb{Z}_\ell)$  is equipped with a bilinear form, symmetric for  $k$  even and alternating for  $k$  odd, with values in  $\mathbb{Z}_\ell(-k)$ . The form induced by tensoring with  $\mathbb{Q}_\ell$  is nondegenerate.

If  $F$  is an lcc  $\mathbb{Q}_\ell$ -sheaf over a smooth scheme  $X$  of pure dimension  $n$  over an algebraically closed field  $k$ , then Poincaré duality gives

$$H^i(X, F)^\vee \simeq H_c^{2n-i}(X, \mathrm{Hom}(F, \mathbb{Q}_\ell(n))),$$

$$H_c^i(X, F)^\vee \simeq H^{2n-i}(X, \mathrm{Hom}(F, \mathbb{Q}_\ell(n))),$$

$$\tilde{H}^i(X, F)^\vee \simeq \tilde{H}^{2n-i}(X, \mathrm{Hom}(F, \mathbb{Q}_\ell(n))).$$

Taking  $X = \bar{M}_n$  and  $F = \mathrm{Sym}^k(R^1 f_{n*} \mathbb{Q}_\ell)$  in these considerations, one defines a nondegenerate bilinear form  ${}_n(\cdot, \cdot)$  on  ${}^k_n W_\ell$  with values in  $\mathbb{Q}_\ell(-k-1)$ . This form is symmetric for  $k$  odd and alternating for  $k$  even. This is the  $\ell$ -adic analogue of the Peterson inner product. If  $n \mid m$ , and if the covering  $\psi : M_m \rightarrow M_n$  is of degree  $d$ , then one has

$${}_m(\psi^* x, \psi^* y) = d \cdot {}_n(x, y).$$

## 1 The Congruence Formula

In this section we fix integers  $k \geq 0$  and  $n \geq 3$ , and prime numbers  $p$  and  $\ell$ . We suppose that  $p$  is prime to  $\ell$  and  $n$ . We denote by  $f_n : E \rightarrow M_n$  the universal elliptic curve over  $M_n$ , equipped with  $\alpha : E_n \xrightarrow{\sim} \mathbb{Z}/(n)^2$ .

Whatever the scheme  $Y$ , one denotes by  $a$  the unique morphism of  $Y$  to  $\mathrm{Spec}(\mathbb{Z})$  or, if necessary, to a subscheme of  $\mathrm{Spec}(\mathbb{Z})$ . If  $Y$  is separated and of finite type over  $\mathrm{Spec}(\mathbb{Z})$ , and if  $F$  is a  $\mathbb{Z}_\ell$ - or  $\mathbb{Q}_\ell$ -sheaf on  $Y$ , one denotes by  $R^i a_*(Y, F)$ , respectively  $R^i a_!(Y, F)$ , respectively  $R^i \tilde{a}(Y, F)$ , the  $\mathbb{Z}_\ell$ - or  $\mathbb{Q}_\ell$ -sheaf over  $\mathrm{Spec}(\mathbb{Z})$  that is the  $i$ th higher direct image of  $F$  under  $a$ , respectively the  $i$ th direct image with proper supports, respectively  $\mathrm{Im}(R^i a_!(Y, F) \rightarrow R^i a_*(Y, F))$ . For  $m \in \mathbb{N}$ , one puts

$$Y[1/m] = Y \times \mathrm{Spec}(\mathbb{Z}[1/m]).$$

**Theorem (4.1, Igusa [1]).** *The scheme  $M_n$  can be compactified to a scheme of curves  $M_n^*$  that is projective and smooth over  $\mathrm{Spec}(\mathbb{Z}[1/n])$ , such that  $M_n^* \setminus M_n$  is an étale covering of  $\mathrm{Spec}(\mathbb{Z}[1/n])$ .*

The schemes  $M_n$  are formally smooth, thus smooth over  $\mathrm{Spec}(\mathbb{Z})$ . The modular invariant  $j$  of the universal curve over  $M_n$  is a morphism of  $M_n$  to the affine line  $\mathbb{A}^1$  over  $\mathrm{Spec}(\mathbb{Z}[1/n])$ . The morphism  $j$  is finite, and is an étale covering away from the sections 0 and 1728 of  $\mathbb{A}^1$ ; in fact:

- (a) Two elliptic curves over an algebraically closed field with the same  $j$ -invariant are isomorphic (e.g. [8] 6.3), thus the geometric fibers of  $j$  are finite. The schemes  $M_n$  and  $\mathbb{A}^1$  being smooth of the same dimension over  $\mathrm{Spec}(\mathbb{Z})$ ,  $j$  is quasi-finite and flat.
- (b) If  $E$  is an elliptic curve over the field of fractions  $K$  of a discrete valuation ring  $R$ , of  $j$ -invariant in  $R$ , and whose points of order  $n$  are rational over  $K$ , then  $E$  has good reduction. The valuative criterion for properness shows thus that  $j$  is proper.

- (c) If  $E$  and  $F$  are two elliptic curves over a scheme  $S$ , of the same  $j$ -invariant, and if  $j$  and  $j - 1728$  are invertible, then the scheme  $\text{Isom}(S; E, F)$  of isomorphisms between  $E$  and  $F$  is étale over  $S$  ([8] 6.3). In the diagram

$$\begin{array}{ccc} \text{Isom}(M_n \times_{\mathbb{A}^1} M_n; \text{pr}_1^* E, \text{pr}_2^* E) & \sim & M_n \times \text{GL}_2(\mathbb{Z}/(n)) \\ \downarrow u & & \downarrow v \\ M_n \times_{\mathbb{A}^1} M_n & \xrightarrow{\text{pr}_1} & M_n, \end{array}$$

where  $j \neq 0, 1728$ ,  $u$  and  $v$  are étale surjective, thus  $\text{pr}_1$  is étale and, by flat descent,  $j$  is étale.

The section at infinity of the projective line  $\mathbb{P}^1 \supset \mathbb{A}^1$  over  $\text{Spec}(\mathbb{Z}[1/n])$  is a regular divisor whose generic point is of characteristic 0 in a regular scheme. It follows from a theorem of Abhyankhar (see [5]) that along this divisor  $j = \infty$ ,  $M_n$  is moderately (tamely?) ramified over  $\mathbb{P}^1$ , and that the normalization  $M_n^*$  of  $\mathbb{P}^1$  in  $M_n$  satisfies (4.1).  $\square$

It follows from the same theorem that the lcc  $\mathbb{Z}_\ell$ -sheaves  $R^i f_{n*} \mathbb{Z}_\ell$  on  $M_n[1/\ell]$  are moderately ramified at infinity. Hence, from (4.1) and the specialization theorems for  $\ell$ -adic cohomology (see [5]), it follows that  $R^i a_*(M_n, \text{Sym}^k(R^1 f_{n*} \mathbb{Z}_\ell))$ ,  $R^i a_!(M_n, \text{Sym}^k(R^1 f_{n*} \mathbb{Z}_\ell))$ , and therefore  $R^i \tilde{a}(M_n, \text{Sym}^k(R^1 f_{n*} \mathbb{Z}_\ell))$  are lcc  $\mathbb{Z}_\ell$ -sheaves over  $\text{Spec}(\mathbb{Z}[1/n, 1/\ell])$ , whose formation is compatible with all base changes.

**Proposition (4.2, Corollary).** *The Galois module  ${}_n W_\ell$  is the fiber of the lcc  $\mathbb{Q}_\ell$ -sheaf*

$$R^1 \tilde{a}(M_n, \text{Sym}^k(R^1 f_{n*} \mathbb{Z}_\ell)) \otimes \mathbb{Q}_\ell$$

*over  $\text{Spec}(\mathbb{Z}[1/n, 1/\ell])$  at the geometric point  $\overline{\mathbb{Q}}$ . It is unramified away from  $n$  and  $\ell$ .*  $\square$

Let the two commutative diagrams over  $M_n \otimes \mathbb{F}_p$

$$\begin{array}{ccc} & \mathbb{Z}/(n)^2 & \\ \alpha \nearrow & & \nwarrow \alpha^{(p)} \\ E_n & \xrightarrow{F} & E_n^{(p)} \\ \cap & & \cap \\ E & \xrightarrow{F} & E^{(p)} \end{array} \quad \begin{array}{ccc} & \mathbb{Z}/(n)^2 & \\ p\alpha^{(p)} \nearrow & & \nwarrow \alpha \\ E_n^{(p)} & \xrightarrow{V} & E_n \\ \cap & & \cap \\ E^{(p)} & \xrightarrow{V} & E \end{array}$$

be written more briefly

$$F : (E, \alpha) \rightarrow (E^{(p)}, \alpha^{(p)}) \quad \text{and} \quad V : (E^{(p)}, p\alpha^{(p)}) \rightarrow (E, \alpha),$$

where  $F$  is the Frobenius morphism and  $V$ , its transpose, is the Verschiebung. These diagrams define morphisms  $\Phi_1$  and  $\Phi_2$  of  $M_n \otimes \mathbb{F}_p$  to  $M_{n,p}$ . These morphisms are finite, considered as sections of  $q_1$  and  $q_2$ , and define a morphism

$$\Phi = \Phi_1 \coprod \Phi_2 : M_n \otimes \mathbb{F}_p \coprod M_n \otimes \mathbb{F}_p \rightarrow M_{n,p} \otimes \mathbb{F}_p.$$

Let  $\Phi^h$  be the restriction of  $\Phi$  to the opens  $M_n^h$  and  $M_{n,p}^h$  of  $M_n \otimes \mathbb{F}_p$  and  $M_{n,p} \otimes \mathbb{F}_p$  corresponding to curves of nonzero Hasse invariant  $h$ .

**Proposition (4.3).**  $\Phi^h$  is an isomorphism.

Let  $\varphi : E_1 \rightarrow E_2$  be a  $p$ -isogeny between elliptic curves of invertible Hasse invariant over a scheme  $S$  of characteristic  $p$ . At each geometric point of  $S$ , either the kernel  $\ker(\varphi)$  of  $\varphi$  is étale over  $S$ , or its Cartier dual, isomorphic to  $\ker({}^t\varphi)$ , is étale over  $S$ . The property “ $\ker(\varphi)$  is étale” is an open property, so that locally on  $S$ , either  $\ker(\varphi)$  is purely infinitesimal, or  $\ker({}^t\varphi)$  is. The only infinitesimal subgroup of order  $p$  of  $E_1$  or  $E_2$  being the kernel of Frobenius, in the first case  $\varphi$  is isomorphic to  $F : E_1 \rightarrow E_1^{(p)}$ , and in the second case  ${}^t\varphi$  is isomorphic to  $F : E_2 \rightarrow E_2^{(p)}$  and  $\varphi$  to  $V : E_2^{(p)} \rightarrow E_2$ .  $\square$

**Proposition (4.4).**

- (i) The scheme  $M_{n,p}$  is smooth over  $\text{Spec}(\mathbb{Z})$  away from points of characteristic  $p$  where  $h = 0$ .
- (ii) The morphisms  $q_1$  and  $q_2$  induce finite flat morphisms  $q'_1$  and  $q'_2$  from the normalization  $M'_{n,p}$  of  $M_{n,p}$  to  $M_n$ .
- (iii) The morphism  $\Phi$  can be factored through a surjective morphism

$$\Phi' : M_n \otimes \mathbb{F}_p \coprod M_n \otimes \mathbb{F}_p \rightarrow M'_{n,p} \otimes \mathbb{F}_p.$$

The automorphism  $\sigma$  of (3.15) exchanges  $\varphi$  and  ${}^t\varphi$ , so that it suffices to prove (i) at the points of characteristic  $p$  of  $M_{n,p}$  where the kernel of  $\varphi$  is infinitesimal: there is no obstruction to infinitesimally lifting an elliptic curve and the infinitesimal part of the kernel of multiplication by  $p$ .

Where  $p = h = 0$ , the fiber of the finite morphism (3.15)  $q_i : M_{n,p} \rightarrow M_n$  is reduced to a point, so that the open of smoothness of  $M_{n,p}$  is dense in  $M_{n,p}$  and  $M'_{n,p}$  is everywhere of dimension 2. The scheme  $M_n$  being regular, following EGA 0<sub>IV</sub> 16.5.1 and 17.3.5(ii), the morphism  $q_i : M'_{n,p} \rightarrow M_n$  is flat. Finally, (iii) results from the fact that  $\Phi$  is finite and  $M_n \otimes \mathbb{F}_p$  is a normal curve.  $\square$

The Hecke endomorphism  $T_p$  of  ${}_nW_\ell$ , such as is made explicit in (3.18), is the tensorization with  $\mathbb{Q}_\ell$  of the fiber at the geometric point  $\overline{\mathbb{Q}}$  of  $\text{Spec}(\mathbb{Z}[1/n, 1/\ell])$  of the endomorphism, again denoted by  $T_p$ , of  $R^1\tilde{a}(M_n, \text{Sym}^k(R^1f_{n*}\mathbb{Z}_\ell))$  defined by the correspondence

$$\begin{array}{ccccc} q_1'^*E & \xrightarrow{\varphi} & q_2'^*E & & \\ \swarrow & & \searrow & & \\ E & \xleftarrow{u} & M'_{n,p} & \xrightarrow{v} & E \\ \downarrow f_n & & \downarrow q_1' \downarrow q_2' & & \downarrow f_n \\ M_n & & M_n & & M_n, \end{array} \quad (4.5)$$

with

$$T_p = q_1'^* \varphi^* q_2'^* \quad (\text{cf. 3.18}).$$

The endomorphisms  $R_p$  and  $I_p$  can be interpreted in the same way.

**Lemma (4.6).** *Let  $X, Y, Z_1$ , and  $Z_2$  be four schemes that are separated and of finite type over a noetherian scheme  $S$ , let  $F$  be a  $\mathbb{Z}_\ell$ -sheaf on  $X$ , let  $G$  be a  $\mathbb{Z}_\ell$ -sheaf on  $Y$ , and denote by  $a$  each of the structural maps from  $X, Y, Z_1$  or  $Z_2$  to  $S$ . Suppose we have the following commutative diagram of  $S$ -schemes and morphisms of sheaves:*

$$\begin{array}{ccccc} y_1^*G & \xrightarrow{z_1} & x_1^*F & y_2^*G & \xrightarrow{z_2} & x_2^*F \\ Z_1 & \xrightarrow{f} & Z_2 & Z_1 & \xrightarrow{f} & Z_2 \\ \downarrow x_1 & & \downarrow x_2 & \downarrow y_1 & & \downarrow y_2 \\ X & & X & Y & & Y \end{array}$$

Suppose that  $f^*z_2 = z_1$ , that  $y_1$  and  $y_2$  are proper, that  $x_1$  and  $x_2$  are finite and flat, and that, for every geometric point  $s$  of  $Z_2$ , the multiplicity of  $s$  in its fiber  $x_2^{-1}(x_2(s))$  is equal to the sum of the multiplicities in their fibers, for  $x_1$ , of the geometric points of  $Z_1$  in the inverse image of  $s$  under  $f$ . Then the diagram

$$\begin{array}{ccccccc} R^i\tilde{a}(Y, G) & \xrightarrow{y_1^*} & R^i\tilde{a}(Z_1, G) & \xrightarrow{z_1} & R^i\tilde{a}(Z_1, F) & \xrightarrow{x_1^*} & R^i\tilde{a}(X, F) \\ \parallel & & \uparrow f^* & & \uparrow f^* & & \parallel \\ R^i\tilde{a}(Y, G) & \xrightarrow{y_2^*} & R^i\tilde{a}(Z_2, G) & \xrightarrow{z_2} & R^i\tilde{a}(Z_2, F) & \xrightarrow{x_2^*} & R^i\tilde{a}(X, F) \end{array}$$

commutes.

This lemma results from the analogous lemmas for  $R^ia_!$  and  $R^ia_*$ . The commutativity of the first squares is trivial. The last is rewritten

$$\begin{array}{ccc} R^i\tilde{a}(Z_1, x_1^*F) & \xleftarrow{\sim} & R^i\tilde{a}(X, x_{1*}x_1^*F) \xrightarrow{\text{Tr}} R^i\tilde{a}(X, F) \\ \uparrow & & \uparrow \parallel \\ R^i\tilde{a}(Z_2, x_2^*F) & \xleftarrow{\sim} & R^i\tilde{a}(X, x_{2*}x_2^*F) \xrightarrow{\text{Tr}} R^i\tilde{a}(X, F), \end{array}$$

and one recalls the definition of the trace in verifying that the square

$$\begin{array}{ccc} x_{1*}x_1^*F & \xrightarrow{\text{Tr}} & F \\ \uparrow & & \parallel \\ x_{2*}x_2^*F & \xrightarrow{\text{Tr}} & F \end{array}$$

is commutative.  $\square$

(4.7) One denotes by  $T_p/\mathbb{F}_p$  the endomorphism induced by  $T_p$  on the restriction to  $\text{Spec}(\mathbb{F}_p)$  of the lcc  $\mathbb{Z}_\ell$ -sheaf  $R^1\tilde{a}(M_n, \text{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell)$ . One has

$$R^1\tilde{a}(M_n, \text{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell)|_{\text{Spec}(\mathbb{F}_p)} \xrightarrow{\sim} R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell);$$

the formation of the trace morphism for a finite flat morphism is compatible with change of base, so that  $T_p/\mathbb{F}_p$  can be constructed, over the model (3.18), starting from the fiber over  $\mathbb{F}_p$  of the correspondence (4.5). Lemma (4.6), applied to the commutative diagrams

$$\begin{array}{ccc} M_n \otimes \mathbb{F}_p \amalg M_n \otimes \mathbb{F}_p & \xrightarrow{\Phi'} & M'_{n,p} \otimes \mathbb{F}_p \\ \downarrow & \swarrow q'_1 & \downarrow \\ M_n \otimes \mathbb{F}_p & & M_n \otimes \mathbb{F}_p \end{array} \quad \begin{array}{ccc} M_n \otimes \mathbb{F}_p \amalg M_n \otimes \mathbb{F}_p & \xrightarrow{\Phi'} & M'_{n,p} \otimes \mathbb{F}_p \\ & \searrow q'_2 & \downarrow \\ & & M_n \otimes \mathbb{F}_p \end{array}$$

furnishes then a decomposition of  $T_p/\mathbb{F}_p$  into the sum of endomorphisms defined by the two following correspondences:

$$(a) \quad \begin{array}{ccc} (E, \alpha) & \xrightarrow{F} & (E^{(p)}, \alpha^{(p)}) = F^*(E, \alpha) \\ \swarrow & & \searrow \\ (E, \alpha) & M_n \otimes \mathbb{F}_p & (E, \alpha) \\ \searrow & & \swarrow F \\ M_n \otimes \mathbb{F}_p & & M_n \otimes \mathbb{F}_p \end{array}$$

where  $F$  is the absolute Frobenius. One recognizes in this correspondence the geometric Frobenius

$$(b) \quad \begin{array}{ccc} x^*(E, \alpha) = (E^{(p)}, p\alpha^{(p)}) & \xrightarrow{V} & (E, \alpha) \\ \swarrow & & \searrow \\ (E, \alpha) & M_n \otimes \mathbb{F}_p & (E, \alpha) \\ \searrow x & & \swarrow \\ M_n \otimes \mathbb{F}_p & & M_n \otimes \mathbb{F}_p \end{array}$$

The map  $x$  is the composite map  $I_p^{-1} \circ F$ :

$$\begin{array}{ccccc} (E, \alpha) & \longleftarrow & (E, p\alpha) & \longleftarrow & (E^{(p)}, p\alpha^{(p)}) \\ \downarrow & & \downarrow & & \downarrow \\ M_n \otimes \mathbb{F}_p & \xleftarrow{I_p^{-1}} & M_n \otimes \mathbb{F}_p & \xleftarrow{F} & M_n \otimes \mathbb{F}_p \end{array}$$

The corresponding endomorphism is thus the composite of

$$V : R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)) \xrightarrow{V^*} R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k(R^1 f_{n*}^{(p)}\mathbb{Z}_\ell)) \xrightarrow{\text{Tr}_F} R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$$

and of

$$I_p^* = \text{Tr}_{I_p^{-1}} : \text{endomorphism of } R^1\tilde{a}(M_n \otimes \mathbb{F}_p, \text{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell).$$

**Proposition (4.8).** *One has  $T_p/\mathbb{F}_p = F + I_p^*V$ , and:*

- (i)  $F$  can be identified with the inverse of the arithmetic Frobenius element  $\varphi_p$  of the Galois group  $\text{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$  acting on  $\tilde{H}^1(M_n \otimes \overline{\mathbb{F}}_p, \text{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$ ;
- (ii)  $F$  and  $V$  are transposes of one another, relative to the scalar product (3.20);
- (iii)  $FV = VF = p^{k+1}$ .

For the relation (i) between the geometric and arithmetic Frobenius, one is referred to the exposé of C. Houzel (SGA 5.XV). The composite  $VF$  is the composite of the homomorphisms obtained from the following maps:

$$\begin{array}{ccccccc} E & \xleftarrow{F_E} & E^{(p)} & \xrightarrow{V_E} & E \\ \downarrow & & \downarrow & & \downarrow \\ M_n & \xrightarrow{F} & M_n & \xrightarrow{F} & M_n \end{array}$$

Thus

$$VF = \text{Tr}_F \circ F_E^* \circ V_E^* \circ F^*.$$

The map  $F_E^* V_E^* = (F_E V_E)^* = (p \cdot 1_E)^*$  acts by multiplication by  $p^k$  on  $\mathrm{Sym}^k R^1 f_{n*} \mathbb{Z}_\ell$ , so that

$$VF = p^k \mathrm{Tr}_F \circ F^* = p^k \cdot p = p^{k+1},$$

because  $F : M_n \rightarrow M_n$  is of degree  $p$ .

By transport of structure,  $\varphi_p$  respects the scalar product (3.20) with values in  $\mathbb{Q}_\ell(-k-1)$ , a group on which  $\varphi_p$  acts by multiplication by  $p^{-k-1}$ . One has thus

$$(Fx, y) = p^{k+1}(\varphi_p Fx, \varphi_p y) = (x, p^{k+1} F^{-1} y) = (x, Vy).$$

□

The following theorem, synonymous with (4.8), goes back to Eichler.

**Theorem** (4.9, The Congruence Formula). *Let  $K_{n,\ell}$  be the largest subextension of  $\overline{\mathbb{Q}}$  that is unramified away from  $n$  and  $\ell$ , let  $\varphi_p$  be a Frobenius element relative to  $p$  in  $\mathrm{Gal}(K_{n,\ell}/\mathbb{Q})$ , let  $F$  be the endomorphism  $\varphi_p^{-1}$  of  ${}_n W_\ell$ , and let  $V$  be the transpose of  $F$  relative to the scalar product (3.20). Then*

$$T_p = F + I_p^* V, \quad FV = p^{k+1}, \quad 1 - T_p X + p R_p X^2 = (1 - FX)(1 - I_p^* V X).$$

□

## 2 Weil implies Ramanujan

If  $p$  is a prime number and  $X$  a scheme over  $\mathbb{F}_p$ , then one denotes by  $\overline{\mathbb{F}}_p$  an algebraic closure of  $\mathbb{F}_p$ , by  $F : X \rightarrow X$  the geometric Frobenius endomorphism, and one puts  $\overline{X} = X \otimes_{\mathbb{F}_p} \overline{\mathbb{F}}_p$ ;  $\ell$  always denotes a prime number different from  $p$ .

By the Weil conjectures one means to claim the following: let  $X$  be a projective smooth scheme over  $\mathbb{F}_p$  and let  $\ell$  be a prime number different from  $p$ . Then the eigenvalues of the endomorphism  $F^*$  of  $H^i(\overline{X}, \mathbb{Q}_\ell)$  are algebraic integers all of whose complex conjugates have absolute value  $p^{i/2}$ .

With the hypotheses and notations of (4.9), one has, recalling that  $(p, n) = 1$ :

**Theorem** (5.1). *If the Weil conjectures are true, then the absolute values of the endomorphism  $F$  of  ${}_n W_\ell$  are algebraic integers, all of whose complex conjugates are of absolute value  $p^{k+1/2}$ .*

Assume the Weil conjectures.

**Lemma** (5.2, modulo Weil). *Let  $X$  be a smooth scheme over  $\mathbb{F}_p$ , which can be represented as an open in a smooth projective scheme  $X^*$ . Then the absolute values of the endomorphism  $F^*$  of  $\tilde{H}^i(X, \mathbb{Q}_\ell)$  are algebraic integers of absolute value  $p^{i/2}$ .*

The natural map of  $H_c^i(X, \mathbb{Q}_\ell)$  to  $H^i(X, \mathbb{Q}_\ell)$  can be factored through  $H^i(X^*, \mathbb{Q}_\ell)$ :

$$H_c^i(X, \mathbb{Q}_\ell) \rightarrow H^i(X^*, \mathbb{Q}_\ell) \rightarrow H^i(X, \mathbb{Q}_\ell),$$

so that, as a  $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ -module,  $\tilde{H}^i(X, \mathbb{Q}_\ell)$  is the quotient of a subobject of  $H^i(X^*, \mathbb{Q}_\ell)$ . □

**Lemma** (5.3, modulo Weil). *Let  $S$  be a smooth scheme over  $\mathbb{F}_p$  and  $f : A \rightarrow S$  an abelian scheme over  $S$ . Suppose that  $A$  can be represented as an open in a projective smooth scheme  $A^*$  over  $\mathbb{F}_p$ . Then the geometric Frobenius  $F^*$  of  $\tilde{H}^i(S, R^j f_* \mathbb{Q}_\ell)$  has eigenvalues that are algebraic integers of absolute value  $p^{i+j/2}$ .*

Let  $m$  be an integer  $> 1$ , and consider the Leray spectral sequences

$$E : E_2^{ij} = H^i(\overline{S}, R^j f_* \mathbb{Q}_\ell) \implies H^{i+j}(\overline{A}, \mathbb{Q}_\ell), \quad {}_c E : {}_c E_2^{ij} = H_c^i(\overline{S}, R^j f_* \mathbb{Q}_\ell) \implies H_c^{i+j}(\overline{A}, \mathbb{Q}_\ell).$$

The endomorphism of multiplication by  $m$ ,  $\psi_m = m \cdot 1_A$ , defines endomorphisms of  $E$  and  ${}_c E$  that fit into a commutative diagram

$$\begin{array}{ccc} {}_c E & \longrightarrow & E \\ \psi_m^* \downarrow & & \psi_m^* \downarrow \\ {}_c E & \longrightarrow & E. \end{array}$$

$\psi_m^*$  acts on  $R^j f_* \mathbb{Q}_\ell$  by multiplication by  $m^j$ , so that  $\psi_m^*$  acts by multiplication by  $m^j$  on the terms  ${}_c E_r^{ij}$  and  $E_r^{ij}$  of  ${}_c E$  and  $E$ . The maps  $d_r$  for  $r \geq 2$  commute with  $\psi_m^*$ , and send  $E_r^{ij}$ , respectively  ${}_c E_r^{ij}$ , to  $E_r^{i'j'}$ , respectively  ${}_c E_r^{i'j'}$ , with  $j \neq j'$ . These are nonzero, and  $E_2^{ij}$ , respectively  ${}_c E_2^{ij}$ , can be identified with a subspace of  $H^{i+j}(\bar{A}, \mathbb{Q}_\ell)$ , respectively  $H_c^{i+j}(\bar{A}, \mathbb{Q}_\ell)$ , where  $\psi_m^* = m^j$ . Consequently,  $\tilde{H}^i(S, R^j f_* \mathbb{Q}_\ell)$  is identified, as a Galois module, with a subspace of  $\tilde{H}^{i+j}(A, \mathbb{Q}_\ell)$  where  $\psi_m^* = m^j$ , and one applies (5.2). The trick used here is due to Lieberman.  $\square$

Let  $f_n : E \rightarrow M_n \otimes \mathbb{F}_p$  be the universal elliptic curve over  $M_n \otimes \mathbb{F}_p$ , and let  $f_{n,k} : E_k \rightarrow M_n \otimes \mathbb{F}_p$  be its  $k$ -fold fiber product with itself. The Künneth formula shows that the  $\mathbb{Q}_\ell$ -sheaf  $R^k f_{n,k*} \mathbb{Q}_\ell$  admits as a direct factor the  $k$ -fold tensor power of  $R^1 f_{n*} \mathbb{Q}_\ell$ ; the latter in turn contains as a direct factor the  $\mathbb{Q}_\ell$ -sheaf  $\text{Sym}^k(R^1 f_{n*} \mathbb{Q}_\ell)$ . Theorem 5.1 results thus from (5.3) and the following lemma.

**Lemma (5.4).** *The scheme  $E_k$  is an open in a smooth projective scheme  $E_k^*$  over  $\mathbb{F}_p$ .*

Let  $E^*$  be the Néron minimal model of  $E$  over  $M_n^* \otimes \mathbb{F}_p$  (4.1). The scheme  $E^*$  is smooth and projective over  $\mathbb{F}_p$ . Since  $n \geq 3$  and the points of order  $n$  of  $E$  form a trivial covering of  $M_n \otimes \mathbb{F}_p$ , the Néron model is semistable, case a or  $b_m$  in Néron's classification. In particular, the projection  $f_n : E^* \rightarrow M_n^*$  has only a finite number of nonsmooth points, and at these points  $f_n$  is nondegenerate, presenting an ordinary quadratic singularity.

Let  $E_k^{**}$  be the  $k$ -fold fiber product of  $E^*$  over  $M_n^*$ . To prove (5.4), it suffices to resolve the singularities of  $E_k^{**}$  without touching the open  $E_k$ . We first prove:

**Lemma (5.5).** *Let  $V$  be the subvariety of affine space over a field  $k$ , with coordinates  $(X_i)_{0 \leq i \leq r}$ ,  $(Y_i)_{0 \leq i \leq r}$ , and  $(T_i)_{1 \leq i \leq s}$ , with equation*

$$X_0 Y_0 = X_1 Y_1 = \cdots = X_r Y_r.$$

*Let  $\mathfrak{m}$  be the ideal of  $\mathcal{O}_V$  generated by the monomials obtained from the monomial  $\prod_{i=0}^r X_i^i$  by a permutation of the coordinates which respects the set of pairs  $\{X_i, Y_i\}$ ,  $0 \leq i \leq r$ . Then  $\mathfrak{m} = \mathcal{O}_V$  away from the singular locus of  $V$ , and the variety  $\tilde{V}$  obtained from  $V$  by blowing up  $\mathfrak{m}$  is smooth over  $k$ .*

The singular locus is the place where the four coordinates  $X_i, Y_i, X_j, Y_j$  for  $i \neq j$  vanish. The open affine of  $\tilde{V}$  defined by the element  $\prod_{i=0}^r X_i^i$  of the ideal  $\mathfrak{m}$  of blowing up is the spectrum of the regular ring

$$k[Y_0/X_1, X_0/X_1, X_1/X_2, \dots, X_{r-1}/X_r, X_r, T_1, \dots, T_s]$$

(for the proof, note that  $X_i/X_{i+1} = Y_{i+1}/Y_i$ ), and 5.5 results.  $\square$

One still shows that, locally for the étale topology, the singularities of  $E_k^{**}$  are isomorphic to those of  $V$  for  $r = k - 1$ , and that this allows the definition on  $E_k^{**}$  of an ideal  $\mathfrak{m}$  analogous to the ideal  $\mathfrak{m}$  of (5.5). Blowing up this ideal, one obtains  $E_k^*$ .  $\square$

An approximation to the following theorem has been proven by Ihara [2].

**Theorem (5.6).** *The Weil conjectures imply the Ramanujan conjecture.*

We note first that (5.1) is true for  $n = 1$ , because  ${}_1^k W_\ell$  is the Galois submodule of  ${}_m^k W_\ell$  invariant under  $\text{GL}_2(\mathbb{Z}/(m))$ . On  ${}_1^k W_\ell$ ,  $I_p^*$  induces the identity, and (4.8) reduces to

$$1 - T_p X + p^{k+1} X^2 = (1 - FX)(1 - VX).$$

The endomorphisms  $F$  and  $V$  are transposes of each other, so that

$$\det(1 - FX; {}_1^k W_\ell) = \det(1 - VX; {}_1^k W_\ell).$$

The action of  $T_p$  on  ${}_1^k W_\ell$  is induced by its action on  ${}_1^k W$ , and is compatible with the decomposition of  ${}_1^k W \otimes \mathbb{C}$  into the sum of the space  $S_{k+2}$  of cusp forms relative to  $\text{SL}_2(\mathbb{Z})$  of weight  $k + 2$ , and of the complex conjugate space. From the Hermitian property for  $T_p$ , for the Peterson inner product, and from (3.19), one deduces then that

$$\det(1 - T_p X + p^{k+1} X^2; {}_1^k W_\ell) = \det(1 - T_p X + p^{k+1} X^2; S_{k+2})^2,$$

and

$$\det(1 - T_p X + p^{k+1} X^2; S_{k+2})^2 = \det(1 - FX; {}_1^k W_\ell)^2,$$

so that

$$\det(1 - T_p X + p^{k+1} X^2; S_{k+2}) = \det(1 - FX; {}_1^k W_\ell). \quad (5.7)$$

Resume the notations of section 1 and make  $k = 10$ . By virtue of Hecke's theory and of (3.19), (5.7) can be rewritten

$$H_p(X) = \det(1 - FX; {}_1^{10}W_\ell),$$

and one applies (5.1). □

One verifies in the same way that the Weil conjectures imply Petersson's generalization of the Ramanujan conjecture.

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**Initials.** EGA: *Éléments de géométrie algébrique*, by A. Grothendieck and J. Dieudonné, Publ. Math. I.H.E.S.

SGA: *Séminaire de géométrie algébrique du Bois-Marie*, mimeographed notes by I.H.E.S., in preparation by North-Holland.